



00 · PROJECT VISION

Mycelium

from private disclosure to a collective emotional ecology

Mycelium is a research prototype that asks how **anonymous emotional expression** can become a **perceivable, evolving shared environment**. A person whispers one sentence into a tree hole; a language model interprets it into ecological parameters; a small emotional body grows, joins a living field, and is slowly changed by the others around it.

Its contribution is not any single feature but a **translation chain** – emotional input → AI interpretation → morphological mapping → social simulation → collective emotional ecology. It is deliberately *not* a mood diary, an AI companion, a wellness app, or a diagnosis tool.

Type

Design research · HCI research prototype

Frame

Speculative interface · research through design

Focus

Indirect emotional co-presence

Status

Living prototype · evaluation planned



Ember Drift



Calm Bell



Moony Moth · you



Mint Pip



Quiet Wisp

01 Background

Young adults are more digitally connected than any generation before them, and can still feel emotionally disconnected – especially while moving through a life transition.

Study abroad, relocation, a first job, remote work, rebuilding relationships in a new city – these transitions rarely sever a person's relationships, but they cut the everyday synchrony those relationships relied on: shared routines, time zones, informal presence. Loneliness, here, is not a shortage of people. It is the gap between the connection one wants and the connection one actually feels (U.S. Surgeon General, 2023).

Platforms make this worse in a specific way. Instagram, TikTok, WeChat, and Discord are excellent at content, reactions, group chat, and recommendation – and poor at showing how a group's emotions settle, influence one another, and shift over time. Much emotional influence is indirect, slow, and ambient; it accumulates through atmosphere and repetition rather than arriving as a message. People are not isolated nodes – they live inside a changing emotional environment that current interfaces cannot show them.

Grounding sources: U.S. Surgeon General, *Our Epidemic of Loneliness and Isolation* (2023); Pew Research Center, social-media use among young adults; *World Happiness Report* (2025); Baek et al., "Lonely Individuals Process the World in Idiosyncratic Ways," *Psychological Science* (2023).

02 Problem

Contemporary platforms make expression and contact visible, but do not support the perception of collective emotional connection.

Young adults in transition are rarely cut off from communication – they remain reachable through feeds, group chats, and DMs. The gap is that these infrastructures privilege **content, response, and visibility** over **atmosphere, continuity, and indirect co-presence**. Individual expression is made legible; the slow, collective, ambient ways feelings move through a group are not.

Framed carefully: the problem is not a lack of communication, and not an individual deficit. It is a lack of *perceivable emotional co-presence* – the felt sense of sharing an emotional environment with others. That gap calls for a different representational and relational model, not another social feed.

03 Target Users

18-30 young adults in transition – defined by life stage, not by pathology. Socially reachable, but often lacking emotional continuity.

In transition study abroad · relocation · first job International students, young migrants, recent graduates, early-career and remote workers, and people rebuilding a social world after a move.	Renegotiating relationships across distance & time Relationships being re-formed across distance, time zones, language, identity, work routines, and shifting social roles.	Not a clinical group life stage, not diagnosis The project speaks to an ordinary, widespread condition of transition – never framed as a disorder to be treated.
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Scope note: the work can begin with young adults broadly and later narrow to international students or relocated young professionals as research focuses.

04 Research Questions

RQ · main	How might anonymous emotional input become a perceivable, evolving collective emotional environment ?
RQ2	How can personal emotional expression be represented without becoming public performance ?
RQ3	How can AI act as a rule generator rather than a conversational companion?
RQ4	How can emotional states be mapped into morphology, behaviour, and relation ?
RQ5	How can a system represent loneliness, support, and transformation as time-based ecological states?

The questions stay exploratory rather than solutionist, and leave room for critique: AI misreading, anonymity, care, and ethical limits.

05 Literature

A small set of strong sources grounds the framing; deeper citation is kept for the appendix and paper. Public-health evidence is separated from HCI / design references.

SOURCE	SUPPORTS
Surgeon General (2023) Epidemic of Loneliness	loneliness as perceived disconnection, not a lack of people
Pew · World Happiness Report (2025)	young adults' high platform use & elevated loneliness
Baek et al. (2023) Psychological Science	lonely people process the world idiosyncratically – a lack of shared understanding
Tad Hirsch design research as psychotherapy; automated evaluation	care, ethics, and contestability in emotionally sensitive design research
Giorgia Lupi Data Humanism	emotional, humane data representation over dashboard metrics
Kramer et al. · emotional contagion	affect can spread through networks – cited critically, given its ethics

Cited as grounding, not as measured claims. Emotional-contagion work is used with care to avoid implying the system measures or manipulates feeling.

06 Design Opportunity

Shift the interaction model – from a platform logic to an ecological one.

FROM	TO
direct communication	indirect emotional co-presence
posting & replying	observing & inhabiting a shared field
emotion as a label	emotion as form, behaviour & relation
AI as response generator	AI as ecological rule generator

The opportunity is not another social platform, but a new representational model for collective emotional presence – one that supports observation, reflection, and low-pressure co-presence, and never asks a person to perform vulnerability in public.

07 Design Principles

The project is guided by restraint. Care is designed as an environmental behaviour, not a reassuring message.

Don't diagnose, don't treat No scores, no diagnosis, no therapy claims. Feelings are made perceivable, not assessed.	No companion, no performance AI is not a friend or chatbot; the visitor is never required to perform vulnerability in public.
Emotion through form & behaviour Feeling becomes visible as morphology and motion – indirect, slow, and observable.	Loneliness as a state, not an identity Represented as a time-based ecological condition that can change, not a fixed label on a person.
Care emerges from relations Support arises from system rules (maturity, time, proximity), not a user-triggered consolation button.	Keep ambiguity, show failure Avoid overclaiming emotional accuracy; keep AI uncertainty and error visible enough to be discussable.

08 Biological Inspiration

Mycelial networks are a **metaphor** for indirect connection – a source of language, not a literal biological model.

Fungal networks connect a forest underground; growth, decay, support, and transformation unfold as slow, temporal processes. Borrowed as design language, they offer words for relation, latency, support, and change – and an alternative imagination to the feed. The metaphor is kept poetic and conceptual on purpose: emotions are not reduced to nature, and the biology is never presented as scientific proof.

BIOLOGY (AS INSPIRATION)	INTERACTION BEHAVIOUR
underground network	indirect connection across a shared field
growth & decay	entities emerge, fade, and change over time
mother trees supporting seedlings	mature entities support isolated ones
nutrient exchange	relation leaves traces between bodies

09 The Role of AI

AI is framed as an **ecological rule generator** – not a chatbot, therapist, or emotional authority.

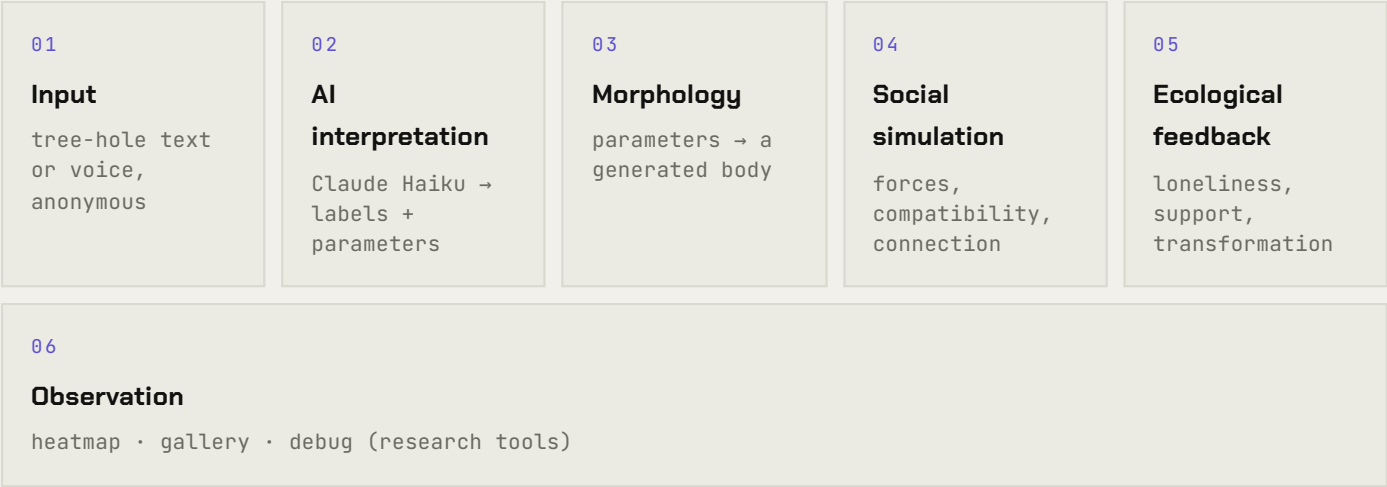
A disclosure is translated into ecological parameters that set the conditions for a new body to emerge. The model creates conditions for emergence rather than generating a comforting reply.

- **Interpretation, not truth.** AI output is treated as a reading that may be wrong – never as a verdict on how you feel.
- **Boundaries are explicit.** Input → interpretation → parameters → organism. The model never speaks back as a persona.
- **Uncertainty is visible.** Error and failed calls are part of the system's story, and future work opens the reading to contestation.

10 System Overview

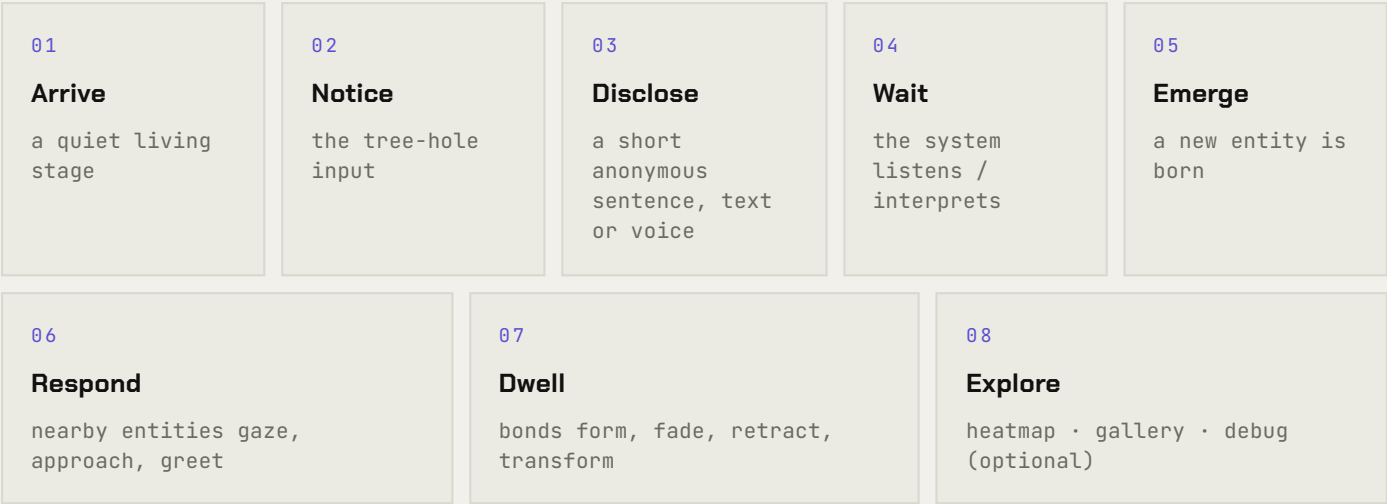
An integrated research prototype in layers – each layer changes the meaning of the last: language becomes data, data

becomes morphology, morphology becomes relation, relation becomes ecology.



11 User Journey

Less like posting, more like releasing something into an environment. The core experience is witnessing your input enter a shared field.



12 Interaction Flow

Simple at the user level, complex at the ecological level. The important interaction is not entering text – it is watching the social field reorganise afterward.

The interface deliberately avoids chat conventions. Greeting, gaze, connection, fading, support, and transformation are all

system responses to a single emotional input.

STEP	USER DOES	SYSTEM RESPONDS
enter	arrives on the stage	ambient field, living population
disclose	writes / speaks into the tree hole	listening / interpreting state
emerge	watches	a new entity is generated
greet	observes	neighbours gaze & approach
evolve	stays	the social field reorganises over time

13 Emotion Pipeline

A translation system, not an emotional diagnosis. Language becomes analysable emotional material, then morphology and behaviour.

INPUT Text / voice one anonymous sentence	MODEL Claude Haiku interprets the disclosure	READING Labels + intensity + topic 18 emotion labels · intensity · topic	PARAMS Morphology six parameters	OUTPUT Entity a unique emotional body
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Error states and uncertainty are part of the narrative: a failed reading is handled gracefully rather than hidden, and – as future work – made contestable.

14 Morphological Mapping

Emotion is represented as morphology rather than text – systematic, but never fully deterministic. The same category still varies through intensity and parameter combinations.

PARAMETER	BECOMES
density	body density · opacity · scale
agitation	jitter · breathing rhythm
tendrilsCount	number of connective tendrils
glow	glow intensity
tintHue	spore colour
particles	spore-emission frequency

FORM

central-axis symmetry +
per-row envelope + edge
dither (density)



COLOUR

tintHue + intensity widen
the palette; local hue
jumps

EXPRESSION

procedural face – the pupil
glances and blinks

RHYTHM

agitation drives jitter & a
breathing cycle

15 Biological Archetypes

Six prototype species form a base vocabulary of emotional bodies – a generative grammar, not fixed emotion mascots.





tender

温柔 · 暖桃

■ H 24°



calm

平静 · 冷蓝

■ H 212°



curious

好奇 · 橙

■ H 32°



dreamy

梦幻 · 薰衣草

■ H 266°



companion

陪伴 · 薄荷

■ H 158°



lonely

孤独 · 冷灰

■ H 220°

Archetypes are starting points: morphology and hybridization mutate them, so the field reads as one system while every member stays distinct. Final species names and the full 18-label set are documented in the appendix.

16 Social Simulation

Entities are not static icons – they participate in a social field. A bond between compatible creatures is shown not as a drawn line but as **tile-swap dyeing**: each takes on the other's palette from the point of contact – a tendril simulated in colour, not stroked as a line.



Contact-point dyeing

the connection, as colour

When two compatible creatures meet, one's cells take on the other's palette from the touch side, sweeping across – the tendril made of pixels. It releases when they part.



Matter exchange

bonded partners trade tiles

Bonded partners ferry small mosaic tiles back and forth – a visible circulation of substance along the (colour) connection.

Compatibility & drift

who bonds with whom

Underneath: attraction, separation, walls, centre-clear, loneliness drift, and a compatibility matrix that decides who bonds – no user control.

BOND LIFECYCLE SHOWN AS

meet	a compatible pair meets; the dye begins sweeping in from the contact side
bonded	held together; the borrowed colour holds ($\leq 45\%$, identity preserved)
part	they drift apart; the colour slowly releases
cooldown	a pause before the pair can dye again

17 Mother Tree



Care as an internal ecological mechanism – not a button, prompt, or chatbot reply.

An entity that has remained on stage long enough (~40s) quietly becomes a **mother tree**. It extends longer-lived, softer support tendrils toward entities that have been socially isolated for a while. Care emerges from maturity, time, and relation inside the field – one of the project's strongest design contributions, precisely because no user has to trigger it.

18 Hybridization

Sustained relation leaves a morphological trace. Encounter becomes visible as memory and mutation.

Cross-type, compatible, sustained contact triggers a state transition through a small state machine – **infection** → **transforming** → **hybrid** – giving relation a temporal arc. Fifteen hybrid forms are implemented.

CONTACT

Infection

compatible pair holds contact

CHANGE

Transforming

form & colour drift toward the other

RESULT

Hybrid

a new, marked body (15 forms)

Terminology under review: "infection" carries clinical / negative connotations. Candidates for public-facing language include *imprint*, *crossing*, *fusion*, or *transformation*.

19 Prototype

A working, feature-rich research artifact – presented as an observable system, not a finished app. Core features carry the research sequence; the rest support it.

ROLE	IMPLEMENTED
Core	living stage · tree-hole input (text + voice) · Claude Haiku emotion analysis (18 labels, intensity, topic) · six morphology parameters · procedural expression (blink, gaze, greeting) · physics field · connection lifecycle (shown as tile-swap dyeing) · compatibility matrix · mother-tree support · loneliness fading · hybridization state machine
Supporting	heatmap of emotional intensity · time-of-day mood / exam mode · hand-gesture interaction · ambient sound & event audio · 15 hybrid forms
Research tools	gallery mode · debug mode (framed as observation instruments, not product features)

Every spore on this page is generated live in your browser by the same deterministic procedure the prototype uses – no hand-drawn art, no image model.

20 Evaluation

The question is not whether the system cures loneliness – it is whether the system makes emotional co-presence **perceivable, interpretable, and discussable**.

At this stage, expert critique and user walkthroughs are more appropriate than product usability testing. Evaluation must carry explicit ethical caution around emotional input and AI interpretation.

METHODS	MEASURES	ETHICS
walkthrough · think-aloud	perceived emotional co-presence	consent · privacy
semi-structured interview	interpretability · trust in AI mediation	emotional sensitivity
expert critique	comfort with anonymity · belonging	no therapy claims
diary · installation observation	awareness of collective emotion · concern about misreading	debriefing

Status: methods planned; an in-product survey is live as one instrument. Findings will be reported qualitatively – this case study makes no claims it has not gathered.

21 Future Work

- **AI contestability.** Let people revise, reject, or reinterpret a reading – "this doesn't feel right" / "reinterpret" / "soften" / "hide."
- **Long-term ecology.** Preserve traces across sessions and community deployments, not just a single visit.
- **Community deployment.** Situated studies with international students, relocated young adults, or a studio cohort.
- **Exhibition mode.** A shared public view that shows the generated field without exposing anyone's original disclosure.
- **Transparent morphology.** Reveal how parameters shape form, and strengthen error / uncertainty design.
- **Expanded evaluation.** Walkthroughs and expert critique before wider deployment.

The most important direction is not more features – it is making the system more transparent, contestable, and situated.

22 Reflection

A mature reflection should make the project more credible, not weaker – limitations become the next research questions.

The work must avoid claiming therapeutic effect. AI interpretation is inherently uncertain and can misread sensitive disclosures; the mycelial metaphor is powerful but risks reading as cute or over-naturalised; and the system is still a prototype whose user research remains to be done. Naming these openly is part of the method, not a retreat from it.

The strongest research identity sits at the intersection of emotional visualization, AI mediation, social simulation, and design ethics – a practice that can build, critique, and responsibly frame emotionally sensitive systems, in the lineage of care-oriented design research (e.g. Hirsch).

A Appendix

A · MORPHOLOGY PARAMETERS

density	body density / opacity / scale
agitation	jitter / breathing rhythm
tendrilsCount	number of tendrils
glow	glow intensity
tintHue	spore colour
particles	spore frequency

B · SYSTEM INVENTORY

18 emotion labels · six morphology parameters · six archetype species · compatibility matrix · connection lifecycle (growing / bonded / retracting / cooldown) · mother-tree support · loneliness fading · hybridization state machine (15 forms) · heatmap · time-mood / exam mode · gesture · ambient audio · gallery · debug.

C · REFERENCE LIST (WORKING)

Surgeon General	Our Epidemic of Loneliness and Isolation (2023)
Pew Research	Social Media Fact Sheet; Teens, Social Media & Technology (2024)
World Happiness Report	2025
Baek et al.	Lonely Individuals Process the World in Idiosyncratic Ways, Psych. Science (2023)
Hirsch	Design research as psychotherapy; automated evaluation & contestability
Lupi	Data Humanism
Kramer et al.	Emotional contagion through social networks (cited critically)

Full citations, the 18-label set, the compatibility matrix, hybrid gallery, and an evaluation-protocol draft are maintained in the project's master document.

MYCELIUM

Design research prototype · from private disclosure to a collective emotional ecology.
Translation chain – emotional input → AI interpretation → morphological mapping → social simulation → collective emotional ecology.
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